Sahaj Bhadja

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Summary

Software engineer with 2+ years supporting **software validation, quality assurance, data controls, risk analysis, change implementation, and technical documentation** across automated, cloud-based, and cross-platform systems.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*

- Implemented data validation, reconciliation checks, error handling, and controlled CI/CD releases across 5,000+ monthly integration jobs, documenting failures and consistently sustaining 99%+ production reliability for cloud-based systems.
- Designed normalized relational schemas with row-level access controls for 500+ users, validating 10,000+ records and reducing dashboard latency from 1.2 seconds to 350 milliseconds through documented optimization.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*

- Established automated testing and release workflows for Python and React services handling 10,000+ daily requests, validating file ingestion, grading outputs, and stable behavior under concurrent production workloads.
- Investigated OCR failures, model hallucinations, and API performance issues, implementing explicit job states, fallback controls, usage logs, and corrective software changes that reduced feedback turnaround 60% for instructors.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Executed reproducible simulation and model-validation workflows across 100,000+ trajectory records, evaluating acceptance metrics and algorithm behavior to improve long-horizon planning 18% and closed-loop stability 15% during testing.
- Maintained experiment configurations, evaluation records, and performance reports using Hydra and Weights & Biases, enabling traceable comparison of architectures and systematic investigation and resolution of unstable model behavior.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Executed cross-platform validation through nightly Jenkins workflows covering installation, integration, and unit tests across Windows, Linux, and macOS, increasing pipeline efficiency 68% and supporting consistent software releases.
- Investigated software defects and integration failures across APIs, AWS Glue, and S3, implementing maintainable corrections that improved response times 30%, reduced payload sizes 25%, and lowered technical debt 20%.

Pandit Deendayal Energy University **India**
Research Fellowship *January 2023 – May 2024*

- Evaluated blockchain consensus mechanisms and five lattice-based cryptographic primitives, identifying security risks and developing a validated algorithm that reduced centralized-control and quantum-risk exposure 25% in benchmark simulations.
- Contributed to a peer-reviewed technical paper and presented findings at IISc Bengaluru, translating complex risk evaluations, design decisions, methods, and measured results into clear technical documentation for review.

Skills

Quality: Software validation, quality assurance, data validation, reconciliation checks, access controls, risk analysis
Lifecycle: Requirements analysis, test planning, unit testing, integration testing, release workflows, issue tracking
Operations: Troubleshooting, failure investigation, error handling, monitoring, performance analysis, CI/CD, automation
Technical: Python, Java, SQL, Shell, PostgreSQL, SQL Server, AWS, Azure, Docker, Linux
Collaboration: Technical documentation, research reports, presentations, Agile, JIRA, Git, cross-functional teamwork

Projects

HMIS - Hospital Management Information System Live | *Data Controls, Monitoring*

- Implemented consent-based patient-history access through ABDM APIs with Appwrite authentication, Sentry monitoring, and structured workflows, securely supporting 100+ daily appointments while reducing record retrieval time by 50%.

Chat2SQL Code | *Data Validation, Change Implementation*

- Designed a schema-aware SQL transformation pipeline with 82% accuracy, implementing query validation and structured execution across relational databases while supporting documented multi-environment changes through Java and TypeScript.

Linguistix | *Automated Validation, CI/CD*

- Developed a containerized Python data-processing API on AWS with automated CI/CD releases and validation checks, achieving 87% measured accuracy and production-grade reliability through repeatable deployment workflows and release checks.